Chapter 6: Renormalization Groups:

Exercise #1: (pg 321)

(from problem) "One-dimensional q-state Potts spin model"

$$\beta H = -K \sum_{i=1}^{N} \delta_{s_i, s_{i+1}} \quad \text{where } s_i = (1, 2, \ldots, q)$$

Fixed points:

$q = 0$ (unstable): Hamiltonian is zero with random spins. because K-coefficient.

$q = 2$ (stable): Hamiltonian has positive values. At low-temperature, H is infinity, same with K-coefficient.

$q > 2$: $\delta_{s_i, s_{i+1}}$ is not regular and a ferromagnet.

Exercise #2: (pg 342)

(pg 337) "Critical exponents"

Symbol	Name	Relationship
α	Temperature	Specific heat $\sim \lvert t \rvert^{-\alpha}$
β	Order Parameter	Magnetic field $\sim -t^{\beta}$
γ	Susceptibility	Susceptibility $\sim \lvert t \rvert^{-\gamma}$

δ	Field Dependence	Magnetic field $\sim \lvert h\rvert^{1/\delta}$
ν	Correlation length	Root distance to Tem $\sim \lvert t\rvert^{-\nu}$
η	correlation function	–

Correlation length:

$$\xi \sim |t|^{-\nu}$$

Scaled: $$b\,\xi = (b^{y_t} \cdot t)^{-\nu}$$

$$1 = -y_t \cdot \nu$$

$$\nu = 1/y$$

Susceptibility:

$$C(r) \sim \begin{cases} r^{2-d-\eta} & r \ll \xi \\ \exp(-r/\xi) & r \gg \xi \end{cases}$$ ...(pg 337)

Scaled: $$b^{y_n} C(r) \sim b^{2-d-\eta} \cdot r^{2-d-\eta}$$

$$y_n = 2-d-\eta$$

$$\eta = 2-d-y_n$$

Correlation Function:

(6.3) "Magnetic Susceptibility"

$$\chi = -\partial_H^2 \cdot F\big|_{H=0}$$

$$= T\,\partial_H^2 \cdot \ln Z\big|_{H=0}$$

$$= \beta \int dr\,dr' \cdot C(r-r')$$

$$\chi = \xi$$

$$\chi = \beta \int d^d r \cdot C(r)$$

$$= \beta \int d^d r \cdot r^{2-d-\eta} \quad \ldots (pg\ 337)$$

$$= \xi^{2-\eta}$$

(pg 337) "correlation length"

$$\xi = |t|^{-\nu}$$

(pg 337) "susceptibility"

$$\chi \sim |t|^{-\gamma}$$

Thus, $|t|^{-\gamma} \sim (|t|^{-\nu})^{2-\eta}$

$$\sim |t|^{-\nu(2-\eta)}$$

$$\gamma = \nu(2-\eta)$$

$$= \frac{2y_h - d}{y_t}$$

<u>Order parameter:</u>

$$M \sim (-t)^{\beta}$$

Scaled: $M \sim (-t)^{(d-y_h)/y_t}$

$$\beta = (d-y_h)/y_t$$

Critical Temperature:

$$C \sim |t|^{-\alpha}$$

Scaled: $C \sim |t|^{-(2y_t - d)/y_t}$

$$\alpha = 2 - d/y_t$$

Field dependence:

$$M \sim |h|^{1/\delta}$$

Scaled: $M \sim |h|^{\frac{d-y_H}{y_H}}$

$$\delta = \frac{y_H}{d - y_H}$$

Exercise #3:
(pg 348)

Parameter:

(6.26) "Action potential- Quadratic"

$$S[\phi] \simeq \int d^d r \left(\frac{r}{2}\phi^2 + \frac{1}{2}(\nabla\phi)^2 - h\phi \right)$$

$$\simeq \int d(ar)^n \left(\frac{r}{2}\phi^2 + \frac{1}{2}(\nabla\phi)^2 - h\phi \right)$$

$$\simeq a^n \int d^n r \left(\frac{r}{2}\phi^2 + \frac{1}{2}(\nabla\phi)^2 - h\phi \right)$$

Propagator lines:

$n^{\underline{th}}$ order - $a^{n-4n/2} = a^{-L}$

n-1 perturbation: $a^{n-4n/2} \circ a^{-1} = a^{-L}$

$$a^{n-4n/2} = a^{-L+1}$$

Exercise #4:
(pg 357)

Pair contraction:

(from problem) "Matrix field"

$$\chi \equiv \langle tr(A_{q,p} \circ W_{p+q} \circ W_{-p}) tr(A_{q',p'} \circ W_{p'+q} \circ W_{-p'}) \rangle$$

where $A, A' \in O(N)$ = matrix fields

$$= \prod_p \big(tr(A_{q,p} \circ W_{p+q}) \circ tr(A_{q',p'} \circ W_{p+q}) - tr(A_{q,p} W_{p+q} \circ A_{q',p'} \cdot W_{p+q}) \big)$$

$$= \prod_p \big(tr(A_{q,p} \circ W_{p+q}) \circ tr(A_{q',p'} \cdot W_{p+q}) - \prod_{p+q} tr(A_{q,p} \circ W_{p+q}) tr(A_{q,p} \cdot W_{p+q}) - tr(A_{q,p} \cdot A_{q',p'}) \big)$$

$$= \prod_p \prod_{p+q} tr(A_{q,p} \circ A_{q',p'})$$

Contractions (1-4)(2-3) & (1-3)(2-4):

(1-4) contracts a tensor, $T_{abcd} = T_{abca}$

(2-3) contracts a tensor, $T_{abcd} = T_{abbd}$

(1-3) Contracts a tensor, $T_{abcd} = T_{abad}$

(2-4) contracts a tensor, $T_{abcd} = T_{abcb}$

Exercise #5:
(pg 359)

(from problem) "Critical exponent"

$$y_h = 2 + \epsilon(N-3)/2(N-2) + O(\epsilon^2)$$

(pg 355) "Dimensions expanded"

$$d = d_c + \epsilon$$

(from problem) "Magnetic field perturbation"

$$h\int d^d r \, tr[g + g^{-1}] = b^d h \int d^d r \, tr[g + g^{-1}]$$

$$= b^{d_c + h} \cdot h \int d^d r \cdot tr[g + g^{-1}]$$

$$= b^{2+\epsilon} \circ h \int d^d r \circ tr[g + g^{-1}]$$

$$\cong b^{y_h} \circ h \int d^d r \circ tr[g + g^{-1}]$$

<u>Susceptibility:</u>

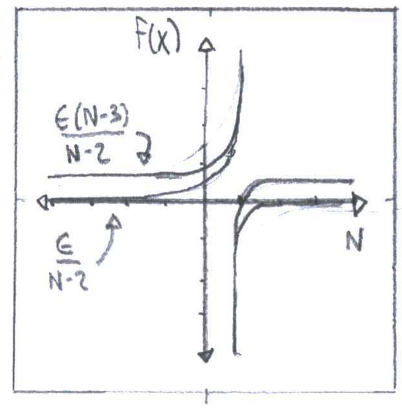
(6.22) "Susceptibility"

$$\eta = 2 + d - 2y_H$$

$$= 2 + d - 2\left(2 + \frac{\epsilon(N-3)}{2(N-2)}\right)$$

$$= \frac{\epsilon(N-3)}{N-2} \qquad \ldots \text{at } d=2$$

$$\cong \frac{\epsilon}{N-2}$$



Exercise #6:

(pg 360)

SU(N) - models:

$$\langle tr(AW_p) tr(A'W_{p'}) \rangle = -\frac{\Pi_p}{2}\left(tr(AA') - \frac{1}{N} tr(AA') \right)$$

(6.34) "SU(N) - completeness relation"

$$\sum_{a=1}^{N^2-1} T^a_{ij} T^a_{kl} = \frac{1}{2}\delta_{il}\cdot\delta_{jk} - \frac{1}{2N}\delta_{ij}\cdot\delta_{kl}$$

$$p_\mu p_{\mu'} \langle tr(\phi_{\mu,q} W_{p+q} W_{-p}) tr(\phi_{\mu'q'} W_{p'+q'} W_{-p}) \rangle_W$$

$$= -p_\mu p_{\mu'} \frac{\Pi_{p+q}}{2}\Big(tr(\phi_{\mu,q} W_{-p}) tr(\phi_{\mu,q} W_{-p})$$

$$- \frac{1}{N} tr(\phi_{\mu,q} W_{-p}) tr(\phi_{\mu,q}\cdot W_{-p'}) \Big)$$

$$= \frac{\Pi_p \Pi_{p+q}}{4} p_\mu p_{\mu'} \circ tr(\phi_{p,-q} \circ \phi_{\mu',-q})$$

RG Equation:

$$\frac{d\ln\lambda}{d\ln b} = \frac{d}{d\ln b} \ln\left(\lambda\left(\frac{(N-2)\lambda}{4\pi} - \epsilon\right)\ln b + \lambda \right)$$

$$= \frac{\lambda\left(\frac{(N-2)\lambda}{4\pi} - \epsilon\right)}{b\left(\lambda\left[\frac{(N-2)\lambda}{4\pi} - \epsilon\right]\ln b + \lambda\right)}$$

$$\simeq \frac{N\lambda}{8\pi} - \epsilon \quad \ldots \text{when } b>1$$

Exercise #7: (pg 364)

(from exercise) "Nonlinear σ-model action"

$$S = \int d^2x \left(\partial n \cdot \partial_n\right)$$

$$= \int d^2x \left((\partial\theta^2) + \sin^2\theta(\partial\phi)^2\right)$$

<u>Topological defect:</u>

$$d\theta = \left(\frac{2r}{1+r^2}\cos\theta, \frac{2r}{1+r^2}\sin\theta\right)$$

$$d\phi = \left(-\sin\phi, \cos\phi\right)$$

$$(d\theta)^2 = \left(\frac{4r^2}{(1+r^2)^2}\cos^2\theta, \frac{4r^2}{(1+r^2)^2}\sin^2\theta\right)$$

$$= \frac{4r^2}{(1+r^2)^2}$$

$$(d\phi)^2 = (1)$$

$$S = \int d^2x \left(\left(\frac{4r^2}{(1+r^2)^2} + \sin^2\theta\right)\right) \ldots \left(\begin{matrix}\text{from}\\ \text{exercise}\end{matrix}\right) \sin\theta = \frac{2r}{1+r^2}$$

$$= 8\int d^2x \, \frac{r^2}{(1+r^2)^2}$$

$$\lim_{r\to\infty} S = \lim_{r\to\infty} 8\int d^2x \, \frac{r^2}{(1+r^2)^2}$$

$$= 0 \quad \text{, "finite"}$$

Exercise #8: (6.45) "Solutions to two-dimensional Poisson Equation"
(pg 366)

$$\psi(r) = 2\pi C(r) \qquad C(r) = \frac{1}{2\pi}\ln(|r|)$$

Exercise #9: (6.47) "Sine-Gordon Model"
(pg 367)

$$S[\theta] = \frac{c}{2}\int d^2r\,(\nabla\theta)^2 + g\int d^2r\cos\theta$$

(pg 322) "Partition function"

$$Z = \int D\theta\, e^{-S[\theta]}$$

$$= \int D\theta\, e^{-\frac{c}{2}\int d^2r\,(\nabla\theta)^2 - g\int d^2r\cos\theta}$$

$$= \int D\theta\, e^{-\frac{c}{2}\int d^2r\,(\nabla\theta)^2} \cdot \sum_{n=0}^{\infty} \frac{(-g)}{n!}\left(\int d^2r\cos\theta\right)^n$$

$$\boxed{\begin{array}{l}\underline{\text{Exponential Series}}: \\ e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}\end{array}}$$

$$= \sum_{n=0}^{\infty} \frac{(-g)}{n!} \int D\theta \, e^{\frac{-c}{2}\int d^2 r (\nabla\theta)^2} \prod_{n=1}^{\infty} \left(\int dr \, \frac{e^{i\theta(r)} + e^{-i\theta(r)}}{2} \right)$$

$$= \sum_{N=0}^{\infty} \frac{y_0^{2N}}{(N!)^2} \prod_{i=1}^{2N} \int d^d r \, e^{-i\sum_{i=1}^{2n} (-1)^i \theta(r)}$$

Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

... when $y_0 = g/2$ and positive, also negative phases appear equally.

Cosine-exponential Relationship:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\simeq \sum_{N=0}^{\infty} \frac{y_0^{2N}}{(N!)^2} \int Dr \, e^{4\pi^2 J \sum_{i<j} C(r_i - r_j)}$$

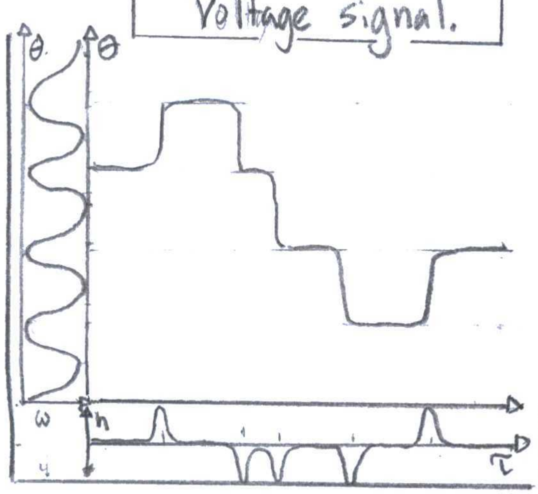
... (from problem) "self-interaction"

$$\langle \theta(r)\theta(r') \rangle = C(|r-r'|)/c$$

$$= 8\pi^2 J \, C(|r-r'|)$$

Problem 6.7.1 - Dissipative Quantum Tunneling - Strong Potential Limit:

A phase (rad) from a voltage signal.



Time of an instanton-event

(6.11) "Action of quantum tunneling"

$$S[\theta] = \frac{1}{4\pi g}\int (d\omega)|\omega||\theta(\omega)|^2 + c\int d\tau \cos(\theta(\tau))$$

(from problem) "Action kinetic energy"

$$S_{kin}[\theta] = (l^2 m/2)\int d\tau\, \dot{\theta}^2$$

a) If $\theta_m = -i f_m \omega_m^{-1} \sum_{i=0} e_i e^{i\omega_m \tau}$, then

$$S_{TOT}[\theta] = S[\theta] + S_{kin}[\theta]$$

$$= \frac{1}{4\pi g}\int (d\omega)|\omega||-i f_m \omega_m^{-1} \sum_{i=0} e_i e^{i\omega_m \tau}|^2$$

$$+ c\int d\tau \cos(\theta(\tau)) + (l^2 m/2)\int d\tau\, \dot{\theta}^2$$

$$= \frac{1}{4\pi g}\int (d\omega)\sum_{ij} e_i(\omega) e_j(\omega) e^{i\omega_m(\tau_i+\tau_j)} \frac{|f(\omega)|^2}{\omega}$$

$$+ n S_{inst}$$

where $S_{inst} = \int d\tau \left(c\cdot\cos(\theta(\tau)) + \frac{l^2 m}{2}\dot{\theta}^2\right)$

(3.71) "Field Integral"

$$Z = \int Dq\, e^{-S[q]}$$

(pg 125) "Free energy"

$$F = -T \ln Z$$

$$= -T \ln\left(\int_{-\infty}^{\infty} Dm\, e^{-\frac{T}{2}\sum_n \int (dq)(\delta + q^2 + \frac{|\omega|}{\Gamma_q})|m_q|^2} \right)$$

$$= \ldots \quad \boxed{\int_0^{\infty} Dm\, e^{-(x+a)^2 m^2} \simeq \frac{1}{(x+a)\sqrt{\pi}} \tanh^{-1}\left(\frac{1}{x+a}\right)}$$

$$\cong 2 \cdot L^d \int dq \tanh^{-1}\left(\frac{\omega / \Gamma_q}{\delta + q^2}\right)$$

$$\simeq 2 \cdot L^d \int dq \int_0^{\Gamma_q} d\omega \coth\left(\frac{\omega\beta}{2}\right) \tanh^{-1}\left(\frac{\omega/\Gamma_q}{\delta + q^2}\right)$$

$\ldots \Gamma\beta \gg 2$

Derivative:

$$\partial_\delta F = \frac{\partial}{\partial \delta}\left[2 \cdot L^d \int dq \int d\omega \coth\left(\frac{\omega\beta}{2}\right) \tanh^{-1}\left(\frac{\omega/\Gamma_q}{\delta + q^2}\right)\right]$$

$$= -2L^d \int dq \int d\omega \coth\left(\frac{\omega\beta}{2}\right) \frac{\omega / \Gamma_q}{(\delta + q)^2 + (\omega/\Gamma_q)^2}$$

b) Renormalization (d_m) and dynamical exponent (z):

(from problem) "Rescale"

$$q' = zq, \quad \omega' = b^z \omega, \quad m_q' = b^{-d_m} \cdot m_q$$

(from problem) "Action for iterant magnetism"

$$= \int Dq\, e^{-\frac{1}{4\pi g}\int dw \sum_{i,j=0} \epsilon_i(w)\epsilon_j(w)\, e^{-iw^2(\tau_i+\tau_j)} \cdot \frac{|f(w)|^2}{|w|}}$$

$\rightarrow -n S_{inst}$

$$= \int\int d\tau\, Dq\, e^{-\frac{1}{4\pi}\int dw \sum_{ij} q(w)^2 \cdot \frac{g|w|}{4|f(w)|^2} \cdot e^{iw(\tau_i+\tau_j)} + i\sum \theta(w) q(w)}$$

$\rightarrow -n S_{inst}$

$$= \int d\tau \cdot e^{-n S_{inst}} \cdot \int Dq\, e^{\frac{-g}{4\pi}\int dw \sum_{i,j} \frac{q(w)^2 |w|}{4 f(w)^2} e^{iw(\tau_i+\tau_j)} + i\sum \theta(w) q(w)}$$

b) $$Z = \int Dq\, e^{\frac{-g}{4\pi}\int (dw) \sum_{i,j} |q(w)|^2 \frac{|w|}{|f(w)|^2}} \cdot \int d\tau\, e^{-n S_{inst} + i\sum \epsilon(w) q(w)}$$

If $\gamma = Z e^{-S_{inst}}$ and $T = \int \frac{1}{f(w)^2} e^{iw(\tau_i+\tau_j)}$

then,

$$Re(Z) = \int Dq\, e^{\frac{-g}{4\pi T}\sum_w |q(w)|^2 |w_m|} \cdot \gamma \int d\tau \frac{\cos(q(\tau))^n}{n!}$$

$$= \int Dq\, e^{\frac{-g}{4\pi T}\sum_w |q(w)|^2 |w_m| + \gamma \int d\tau \cos(q(\tau))}$$

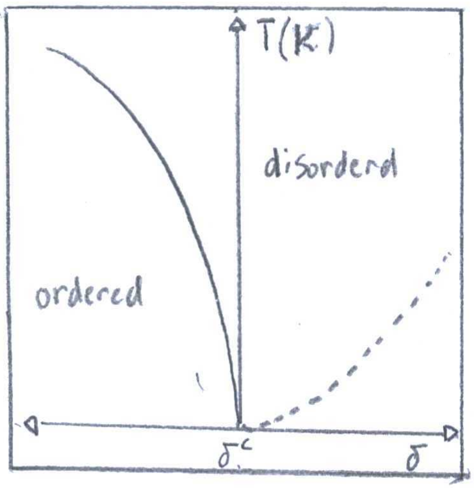
Identities above:

Eulers Formula:	Hubbard-Stratonovich:
$e^{ix} = \cos(x) + i\sin(x)$	$e^{-\frac{a}{2}x^2} = \int dy\, e^{-\frac{y^2}{2a} + ixy}$

Exponential Series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Problem 6.7.2 - Quantum Criticality:



Compare and Contrast:

Critical Exponents	Debye / Einstein Model
◦ Specific heat	◦ Specific heat
◦ Temperature	◦ Temperature
◦ Order parameter	◦ Proportional
◦ Spin dependent	◦ Substance dependent

closer to constants in 2025

(pg 375) "Action for iterant magnetism"

$$S[m] = \frac{T}{2}\sum_n \int (dq)\left(\delta + q^2 + \frac{|\omega_n|}{\Gamma_q}\right)|m_q|^2 + \frac{u}{4}\int dx\, m^4(x)$$

(from problem) "Gaussian theory"

$$\delta > 0 \;,\; u = 0$$

(3.71) "Field integral"

$$Z = \int Dm\, e^{-S[m]}$$

$$= \int Dm\, e^{-\frac{T}{2}\sum_n \int (dq)\left(\delta + q^2 + \frac{|\omega_n|}{\Gamma_q}\right)|m_q|^2}$$

$$S[m] = \frac{T}{2}\sum_n \int (dq)\left(\delta + q^2 + \frac{|w_n|}{\Gamma_q}\right)|m_q|^2 + \frac{u}{4}\int dx\, m^4(x)$$

$$= \frac{1}{2}\int_{\Lambda}^{\Lambda/b}(dq)\int_0^{\Gamma_q}(dw)\left(\delta + q^2 + \frac{|w_n|}{\Gamma_q}\right)|m_q|^2$$

$\circ\circ\circ$ from statement $u=0$, $T\sum_n = \int (dw)$

$$= \frac{b^{-d}}{2}\int_{\Lambda}^{\Lambda/b}(dq)^d\int_0^{\Gamma_q}(dw)\left(\delta + (q'^2 b^{-2}) + \frac{|b^{-z}w|}{\Gamma_q}\right)\circ$$

$$\times |b^{d_m}\circ m_q|^2$$

If $-d-2-z+2d_m=0$

then, $d_m = \dfrac{(d+2+z_1)}{2}$

$$S[m] = \frac{1}{2}\int^{\Lambda/b}(dq)^{d+z}\int(dw)\left(\delta + q^2 + \frac{|w|}{\Gamma_q}\right)|m_q|^2$$

$$= \frac{b^{-d-z}}{2}\int^{\Lambda}(dq)^{d+z}\int(dw)\left(\delta + q\circ b^{-2} + \frac{|w|\cdot b^{-z+1}}{\Gamma_q}\right)|b^{2d_m}||m_q|^2$$

If $-d-z_1-2-z+2d_m=0$

then $z=1$ from previous

equality.

Scaling dimension (y_n):

$$S[m] = \frac{b^{-d-2}}{2}\int^{\Lambda}(dq)^{d+z}\int(d\omega)\left(\delta\cdot b^{2y_\delta}+q^2\cdot b^{-2}+\frac{|\omega|b^{-z}}{\Gamma_q}\right)\cdot$$

$$\times\, b^{2d_m}\cdot|m_q|^2$$

If $-d-2+2y_\delta-2-z+2d_m = -d-2-z+2d_m$

$$y_\delta = 2$$

Quartic Interaction (y_u):

$$S[m] = \frac{T}{2}\sum_n\int(dq)\left(\delta+q^2+\frac{|\omega|}{\Gamma_q}\right)|m_q|^2+\frac{u}{4}\int dx\, m^4(x)$$

$$= \frac{-b^{-d-z}}{2}\int(dq)^{d+z}\int(d\omega)\left(\delta\cdot b^2+q^2\cdot b^{-2}+\frac{|\omega|b^{-2}}{\Gamma_q}\right)b^{2d_m}\cdot|m_q|^2$$

$$\cdot\frac{b^{y_u}\cdot u}{4}\int dx\, b^{4d_m}\cdot m^4(x)$$

If $-d-2-z+2d_m = -d-2-z+2d_m+y_u+4d_m$

$$y_u = -4d_m$$

$$= -2(d+3) \quad @\ z=1$$

c) $$\langle m_f^2\rangle = 2\int_0^1\int_0^1(dq)(d\omega)\coth\left(\frac{\omega\beta}{2}\right)\frac{\omega\Gamma_q}{(\delta+q^2)^2+(\omega/\Gamma_q)^2}$$

$\boxed{\text{Try the integral!}}$

If $\delta' = \delta(1+2\ln b) + 12uf(T)$

$u' = b^{4-d-z}$

$T' = b^z T$

then $\dfrac{dT}{d\ln b} = zT$

$$\frac{du}{d\ln b} = (4-d-z)u$$

$$\frac{d\delta}{d\ln b} = 2\delta + uf(T)$$

d) $\dfrac{dT}{d\ln b} = \dfrac{dT}{du}$ ...if $u = \ln b$

$= zT$

$T(b) = e^{z\ln b + C}$

$= C \cdot e^{z\ln b}$

$= C \cdot b^z$

$= Tb^z$ .. if $C = T$

$\boxed{\text{Linear First order Differential Equation}}$

$$\frac{du}{d\ln b} = (4-d-z)u$$

$$u(b) = u b^{4-d-z} \quad \therefore \text{if } c = u$$

Linear First order Nonhomogeneous Differential

Substitution

$$\frac{d\delta}{d\ln b} = \frac{d\delta}{dx} \quad \therefore \text{if } x = \ln b$$

$$= 2\delta + u f(T)$$

Arrange

$$\frac{d\delta}{dx} - 2\delta = u f(T)$$

Factor

$$\frac{d\delta}{dx} \cdot e^{-2x} - 2\delta \cdot e^{-2x} = u f(T) \cdot e^{-2x}$$

Simplify

$$\frac{d}{dx}\left(\delta \cdot e^{-2x}\right) = u f(T) \cdot e^{-2x}$$

Integrate

$$\delta \cdot e^{-2x} = \int u f(T) \cdot e^{-2x} dx$$

Solve

$$\delta(b) = e^{2x}\left(C + \int u f(T) e^{-2x} dx\right)$$

$$= \delta b^2 + b^2 u \int dx \cdot \frac{f(T)}{b^2}$$

Problem 6.7.3 – Kondo Effect, Poor mans scaling:

a) (6.60) "Schrödinger equation"

$$\left(\hat{H}_{10} \frac{1}{E - \hat{H}_{00}} \hat{H}_{01} + H_{11} + H_{12} \frac{1}{E - H_{22}} H_{21}\right) |\psi_1\rangle = E |\psi_1\rangle$$

(from problem) "sd-Hamiltonian"

$$\hat{H}_{sd} = \sum_{k\sigma} \epsilon_k c^\dagger_{k\sigma} c_{k\sigma} + \sum_{kk'} \Big(J_z \hat{S}^z (c^\dagger_{k\uparrow} c_{k'\uparrow} - c^\dagger_{k\downarrow} c_{k'\downarrow}) + J_+ \hat{S} c^\dagger_{k\downarrow} c_{k'\downarrow} + J_- \hat{S}^- c_{k\uparrow} c_{k'\downarrow} \Big)$$

<u>At band edge:</u>

$$\left(\hat{H}_{12} \circ \frac{1}{E-H_{22}} \circ \hat{H}_{21}\right)|\psi_1\rangle = \sum_{kk'} \Big(J_z \hat{S}^z (c^\dagger_{k\uparrow} c_{k'\uparrow} - c^\dagger_{k\downarrow} c_{k'\downarrow}) + J_+ \hat{S} c^\dagger_{k\downarrow} c_{k'\downarrow} + J_- \hat{S}^- c^\dagger_{k\uparrow} c_{k\downarrow} \Big)$$

$$\times \left(\frac{1}{E-H_{22}} \right) \times$$

$$\sum_{kk'} \Big(J_z \hat{S}^z (c^\dagger_{k\uparrow} c_{k'\uparrow} - c^\dagger_{k\downarrow} c_{k'\downarrow}) + J_+ \hat{S} c^\dagger_{k\downarrow} c_{k'\downarrow} + J_- \hat{S}^- c^\dagger_{k\uparrow} c_{k\downarrow} \Big)$$

$$= J_+ J_- \sum \hat{S}^- c_{k\uparrow} c_{k\downarrow} \frac{1}{E-H_{22}} \hat{S}^+ c_{k\downarrow} c_{k\uparrow} |\psi_1$$

+ multiple terms ...

b) $$J_+ J_- \sum_{k_S, k_S k_F} S^- c^\dagger_{k\uparrow} c_{k\downarrow} \frac{1}{E-H_{22}} \hat{S}^+ c^\dagger_{k\downarrow} c_{k\uparrow}$$

$$= \frac{J_+ J_- \hat{S}^- \hat{S}^+}{E-H_{22}} \sum_{k_S k_S} \int_{D/b}^{V_0 D} dk_F \circ c^\dagger_{k\uparrow} \cdot c_{k\downarrow} \circ c^\dagger_{k\downarrow} \circ c_{k\uparrow}$$

$$= J_+ J_- \sum_{k_s k_f} V_0 D(1-b^{-1}) \hat{S}^- \hat{S}^+ C_{kf}^\dagger C_{k\uparrow} \frac{1}{E - D + \epsilon_k - H_0}$$

$\circ\circ$ when $H_{22} = H_0 + D - \epsilon_k$

$$= J_+ J_- \sum_{k_s k_s} V_0 (1-b^{-1}) \left(\tfrac{1}{2} - S^z\right) C_{k\uparrow}^\dagger C_{k\uparrow}$$

$\circ\circ\circ$ when $\hat{S}^- \hat{S}^+ = 1/2 - S^z$

and $E, H_0, \epsilon \ll 1$